COA MOST EXPECTED QUESTIONS(PYQs)

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NOTE: Numeric values may differ and slight changes occur in numerical but the concept remains the same.

UNIT 1

SHORT QUESTION

Q1List the difference between static RAM and dynamic RAM.
Q2 Define the terms Computer architecture and Computer organization.
Q3 What is mean by bus arbitration? List different types of bus arbitration
Q4Differentiate between Daisy chaining and centralized parallel Arbitration
Q5 What is the bus? Explain the different type of buses.
Q6 What is memory transfer and bus transfer?
Q7 List three types of control signals.

LONG QUESTIONS

Q1 Describe the different kinds of addressing modes with examples and
An instruction is stored at location 400 with its address field at location 401. The address field has the value 500. A processor register R1 contains the number 200. Evaluate the effective address if the addressing mode of the instruction is (i) direct (ii) immediate (iii) relative (iv) register indirect (v) index with R1 as an index register

Or

What is an effective address? How it is calculated in different types of addressing modes? Explain.
Q2 Discuss stack organization. Explain the register stack and memory stack
Q3 Explain the functional units of a computer system in detail
Q4 Draw a diagram of a Bus system in which it uses 3 state buffers and a decoder instead of the multiplexers.
Q5 What is a memory stack? Explain its role in managing subroutine with the help of a neat diagram
Q6 Draw a diagram of a bus system using MUX which has 4 registers of size 4 bits each

Or

Explain the bus transfer through multiplexer

Unit 2

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SHORT QUESTIONS

Q1 Perform the following operation on signed numbers using 2's compliment method: (56)10 + (-27)10
Q2 Perform 2s complement subtraction of smaller number (101011) from (111001)
Q3 What is multiplexer and decoder
Q4 Design a 4-bit combinational incremental circuit using four full adder circuits.

LONG QUESTIONS

Q1 Show the systematic multiplication process of (-15) × (-16) using Booth's Algorithm and also write Booth's algorithm

Or

Show the systematic multiplication process of (-8) × (9) using Booth's Algorithm

Or

Show the systematic multiplication process of (11000) × (01010) using Booth's Algorithm both are unsigned values (here we do not need to convert in binary)

Q2 Explain the IEEE-754 standard for floating point representation. Express $(314.175)_{10}$ and $(-307.1875)_{10}$ in all the IEEE-754 models
Q3 Describe the derivation procedure of the look-ahead carry adder by an example with the help of block diagram (4bit, 3bit any bit you want to draw)

or

Explain in detail the principle of carry look-ahead adder and design a 4-bit CLA adder
Q4 Explain 2-bit by 2-bit Array multiplier. Draw the flowchart for the divide operation of two numbers in signed magnitude form.
Q5 Draw a flowchart of adding and subtracting fixed point binary numbers where negative numbers are signed one's complement
Q6 Draw the Data path of 2's compliment multiplier. Give the Robertson multiplication algorithm for 2's complement fractions. Also, illustrate the algorithm for 2's compliment fraction by a suitable example.
Q7 Describe the Sequential Arithmetic & Logic unit (ALU) using the proper diagram

Unit 3

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SHORT QUESTIONS

Q1 What is micro-operation and micro-instruction
Q2 What is microcode microprogram and control word
Q3 Differentiate between Horizontal & Vertical microprogramming.
Q4 Register A holds the binary value 10011101.What is the register value after the arithmetic shift right? Starting from the initial number 10011101, determine the register value after the arithmetic shift left, and state whether there is an overflow

LONG QUESTIONS

Q1. What is instruction format. Explain all its types. Evaluate the arithmetic statement.
X = A + B * [C * D + E * (F + G)]
using a stack organized computer with zero address operation instruction
(ii)one address (ii) two address(iii) three address
Q2 What is instruction cycle draw the flowchart of instruction cycle
Q3 What is control unit. How many types of control unit? Explain the hardwired control unit in detail. Discuss the methods of designing hardwired control unit
Q4 Explain the organization of microprogrammed control unit.
Q5 What is micro programmed control unit? Give the basic structure of micro programmed control unit. Also discuss the microinstruction format and the control unit organization for a typical micro programmed controllers using suitable diagram
Q6 Explain RISC and CISC Processor. Explain with neat diagram, the address selection for control memory
Q7 What is a microprogram sequencer? With block diagram, explain the working of microprogram sequencer
Q8 Explain the concept of pipelining and also explain types of pipelining
Consider a pipeline having 4 phases with duration 60,50,90 and 80 ns. Given latch (register delay)
Calculate (i) pipeline cycle time (ii) non-pipeline cycle time (iii) speed up ratio (iv) pipeline time and sequential time for 1000 task (v) throughput
Q9 A vertical microprogrammed control unit support 512 instructions. The system is using 8 conditional flags and contain 31 control signals. Each instruction on an average has 1 micro-operations .calculate the approximate size of memory in bytes

Unit 4

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SHORT QUESTIONS

Q1 What is memory hierarchy?
Q2 What is cache hit and miss?
Q3 What is virtual memory?
Q4 What is interleaved memory?
Q5 What is a locality of reference?

LONG QUESTIONS

Q1 Discuss different mapping techniques in cache memory. Explain its merits and demerits
Q2 Give the structure of Commercial 8MX8 bit DRAM Chip
Q3 Write a short note on a magnetic disk, magnetic tape, and optical disk.
Q4 computer uses RAM chips of 1024*1 capacity. i) How many chips are needed & how should their address lines be connected to provide a memory capacity of 1024*8? ii) How many chips are needed to provide a memory capacity of 16 KB
Q5 Calculate the page fault for a given string with the help of LRU & FIFO AND optimal page replacement algorithm, Size of frames = 4 and string 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6
Q6 The logical address space in a computer system consists of 128 segments. Each segment can have up to 32 pages of 4K words each. Physical memory consists of 4K blocks of 4K words each. Formulate the logical and physical address formats.
Q7 How is the Virtual address mapped into physical address? What are the different methods of writing into cache?
Q8 A digital computer bus system for 8 register of 16 bit each. The bus is constructed with multiplexer
1.How many selection input is there in each multiplexer
2. What is the size of multiplexer needed
3.How many multiplexers are there in bus?
Q9 A ROM chip of 1024*8 capacity has four select inputs and operated from a 5-volt power supply. How many pin are needed? Draw the b lock diagram and label all input and output terminal in ROM
Q10 What is associative memory? Explain with the help of a block diagram. Also Mention the situation in which associative memory can be utilized.

OR

What do you mean by CAM? Explain its major characteristics

Q11 A digital computer has a memory unit of 64k x 16 and cache memory of 1k words the cache uses direct mapping with a block size of 4 words

1 How many bits are there in the tag, index (cache line), block and word (block offset)

2 How many blocks the cache accommodate

3 How many bits are there in each word in cache and how they are individual into function include a valid function

Q12 A two-way associative memory uses a block of 4 words A cache can accommodate a total of2048words from memory. the main memory size 128k x32

1. **What is the size of cache memory?**
2. **Formulate the necessary information to construct cache memory(cache line , block offset)**

NOTE

HERE ONLY EXAMPLE OF MAPPING QUESTION IS GIVEN, QUESTION MAY VARY OF MAPPING MAY BE OF FULL ASSOCIATIVE MAY BE DIRECT, MAY DIFFERENT VALUE GIVE BUT CONCEPT REMAIN SAME

Unit 5

SHORT QUESTIONS
Q1 Describe cycle stealing in DMA
Q2 Define the role of MIMD in computer architecture
Q3 What are the function performed by input output unit
Q4 why DMA get priority over CPU when request memory transfer
Q5 Difference between memory mapped and i/o mapped
Q6 what is trap and exception
Q7 What is Flynn's' taxonomy
LONG QUESTIONS
Q1 what is interrupt? Explain different types of interrupt? How interrupts are handled?
Q2 What do you mean by asynchronous data transfer. Explain strobe (source and destination initiated) and hand shaking transfer (source and destination initiated)
Q3 Discuss the design of typical input output interface
Q4 Explain how the computer buses can be used to communicate with memory and I/O. Also draw the block diagram for CPU-IOP communication.
Q5 Discuss different modes of data transfer Or Explain the interrupt driven technique and design its flowchart along with its merits and demerits
Q6 with a neat schematic diagram, explain about DMA controller and its mode of data transfer and write the different modes of data transfer in DMA
Q7 what is IOP processor? What is serial communication

NOTE

<u>Revision cycle suggest to follow</u>

- **Firstly, do Unit1(Theory + addressing modes numerical)**
- **Then Unit2+ (theory + booths and IEEE and instruction numerical must)**
- **After that Unit5 (only theory + DMA MUST)**
- **Then unit 3 + (theory + pipelining and microprogramming numerical)**
- **Finally, unit 4 (theory + cache mapping numerical must)**

ALL THE BEST STUDENTS DO YOUR BEST !